VOICE-BASED CONCEPT UNDERSTANDING ANALYSER

Project Documentation

ACKNOWLEDGEMENT

I express my sincere gratitude to everyone who contributed to the successful completion of the project **Voice-Based Concept Understanding Analyser (VBCUA)**.

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ABSTRACT

Voice-Based Concept Understanding Analyser (VBCUA) is an Artificial Intelligence-based educational application developed to evaluate a student's conceptual understanding through spoken explanations. Traditional assessment methods primarily evaluate written responses and often fail to accurately measure a student's actual understanding and communication skills. This project addresses that limitation by automatically analyzing spoken explanations using modern AI techniques.

The application is developed using **Python** and **Streamlit**, providing an interactive and user-friendly web interface. Users can either record their explanations directly through a microphone or upload pre-recorded audio files. The recorded speech is converted into text using **OpenAI Whisper**, an advanced speech recognition model known for its high transcription accuracy.

After transcription, the generated text is compared with predefined reference concepts using **Sentence-BERT**, which calculates semantic similarity and determines how closely the student's explanation matches the expected concept. The system further evaluates key concept coverage by identifying which important concepts were successfully explained and which were omitted.

In addition to conceptual evaluation, the application performs detailed speech fluency analysis using **Librosa**, extracting various audio features such as speaking rate, pause ratio,

filler words, speech duration, and overall fluency. These metrics are combined with semantic similarity and concept coverage scores to calculate a comprehensive final evaluation score.

The application presents the evaluation results through an interactive dashboard consisting of graphical charts, progress indicators, detailed feedback, transcription results, semantic analysis, fluency metrics, and personalized suggestions for improvement. Finally, a professionally formatted PDF report is automatically generated, allowing students and educators to download and review the complete assessment.

The proposed system provides an efficient, objective, and intelligent approach for evaluating conceptual understanding, making it suitable for educational institutions, online learning platforms, technical interviews, and skill assessment systems.

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PROJECT DESCRIPTION

Voice-Based Concept Understanding Analyser (VBCUA)

Voice-Based Concept Understanding Analyser (VBCUA) is an Artificial Intelligence-powered educational assessment platform that evaluates a learner's conceptual understanding by analyzing spoken explanations. The application combines speech recognition, semantic similarity analysis, natural language processing, and audio feature extraction to provide an objective and comprehensive evaluation of a student's knowledge and communication skills.

The system allows users to either record their explanations through a microphone or upload audio recordings in multiple formats. The speech is accurately transcribed into text using **OpenAI Whisper**, after which **Sentence-BERT** compares the transcription with predefined reference explanations to determine semantic similarity and concept coverage. Simultaneously, **Librosa** analyzes the audio signal to measure speech fluency through parameters such as speaking rate, pause ratio, speech duration, and filler word frequency.

Based on semantic understanding, concept coverage, and fluency analysis, the system calculates an overall performance score and classifies the learner's understanding into different performance levels. The application presents detailed visual feedback through an interactive Streamlit dashboard, allowing users to understand both their strengths and areas requiring improvement.

The application also generates a professional PDF evaluation report containing transcription results, semantic analysis, concept coverage, fluency metrics, overall score, and personalized recommendations. This report can be downloaded and shared for academic evaluation or progress tracking.

Developed using **Python**, **Streamlit**, **OpenAI Whisper**, **Sentence-BERT**, **Librosa**, **Pandas**, and **FPDF2**, VBCUA demonstrates the practical integration of Artificial Intelligence and Machine Learning technologies in modern educational assessment systems.

Scenarios

Scenario 1: Student Preparing for Semester Examination

A student preparing for a semester examination selects the concept **Operating System**, records a spoken explanation through the microphone, and submits it for evaluation. The application transcribes the speech using OpenAI Whisper and compares it with the reference explanation using Sentence-BERT. The system evaluates semantic similarity, identifies covered and missed key concepts, analyzes speech fluency, and generates an overall performance score along with personalized feedback. Finally, the student downloads a PDF report to review strengths and areas for improvement before the examination.

Scenario 2: Faculty Assessing Student Understanding

A faculty member asks students to explain a computer science concept verbally during laboratory sessions. Each student uploads or records their explanation using the VBCUA application. The system automatically evaluates conceptual understanding, measures concept coverage, analyzes speaking fluency, and produces standardized evaluation reports. This significantly reduces manual assessment effort while ensuring fair and objective evaluation for every student.

Scenario 3: Self-Learning and Interview Preparation

A learner preparing for technical interviews practices explaining concepts such as Data Structures, Database Management Systems, or Computer Networks using the application. VBCUA evaluates the explanation, identifies missing concepts, highlights filler words and excessive pauses, and provides constructive suggestions for improvement. Continuous practice enables learners to improve both conceptual understanding and communication skills before interviews.

Technical Architecture

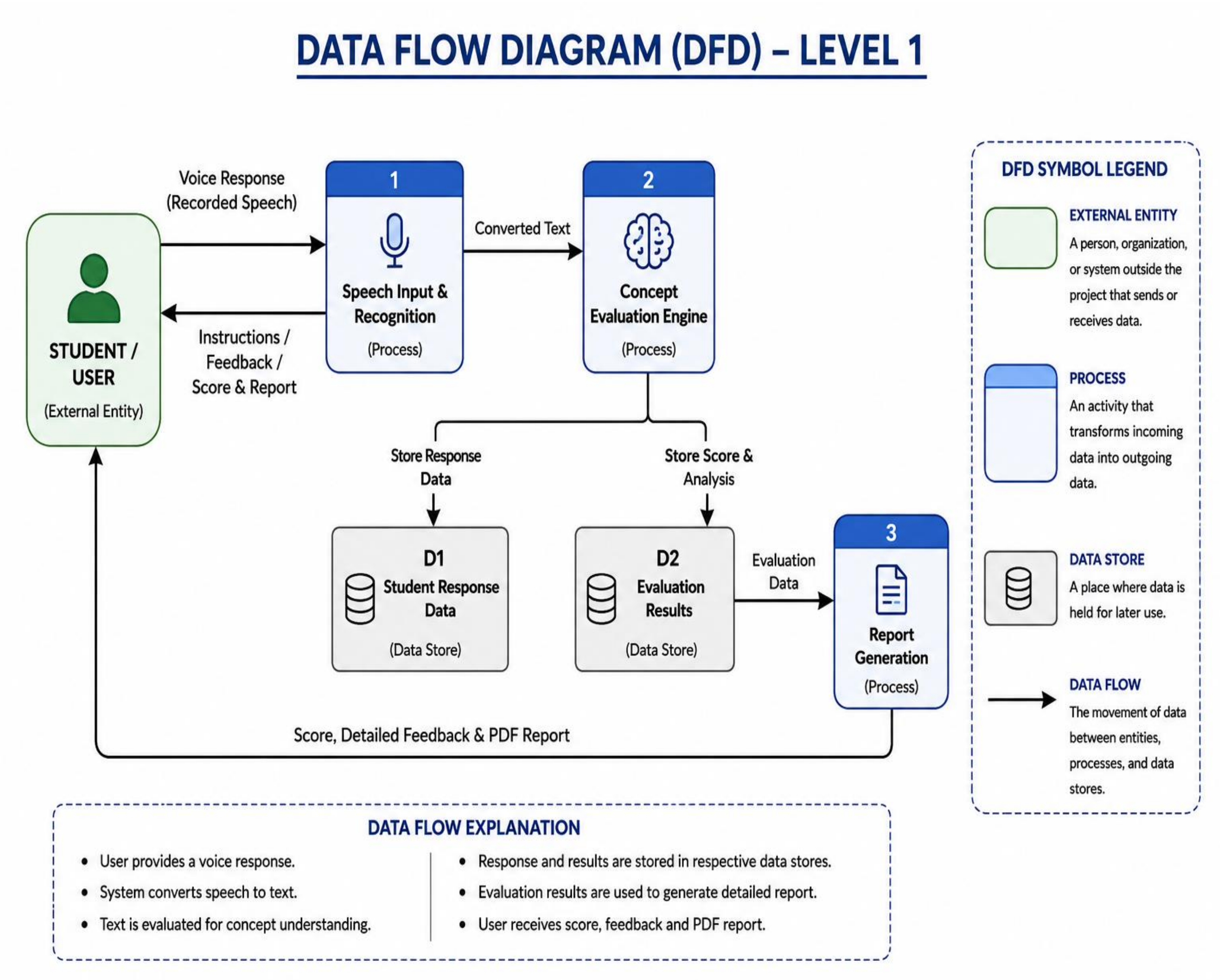
Description

Voice-Based Concept Understanding Analyser (VBCUA) follows a modular Artificial Intelligence architecture that integrates speech recognition, Natural Language Processing (NLP), semantic similarity analysis, audio signal processing, and automated report generation. The application is developed using **Streamlit**, which provides an interactive and responsive web interface. The backend coordinates communication between different AI modules responsible for speech transcription, semantic evaluation, audio feature extraction, scoring, and PDF report generation.

The system begins by receiving an audio explanation from the user either through microphone recording or audio file upload. The audio is processed using **OpenAI Whisper**, which converts speech into accurate text. The transcribed text is then analyzed using **Sentence-BERT**, where semantic similarity between the student's explanation and predefined reference concepts is calculated. Simultaneously, **Librosa** extracts audio features such as speaking rate, pause ratio, speech duration, and filler words to determine speaking fluency.

The semantic similarity score, concept coverage score, and fluency score are combined using a weighted scoring algorithm to generate the final evaluation score. The application displays comprehensive analytical results through an interactive dashboard and generates a downloadable PDF report containing the complete assessment.

System Architecture Diagram



Pre-requisites

Before developing the Voice-Based Concept Understanding Analyser (VBCUA), the following software tools, libraries, and technical knowledge are required.

Programming Knowledge

- Python Programming
- Object-Oriented Programming (OOP)
- Basic Machine Learning Concepts
- Natural Language Processing (NLP)
- Audio Signal Processing Fundamentals

Frameworks & Libraries

- Streamlit

- OpenAI Whisper
- Sentence-Transformers (Sentence-BERT)
- Librosa
- Pandas
- NumPy
- Matplotlib
- NLTK
- FPDF2
- Pydub
- SciPy

Software Requirements

- Python 3.10 or above
- Visual Studio Code
- Git & GitHub
- Streamlit Framework
- Internet Connection (for downloading Whisper model initially)

Hardware Requirements

- Windows/Linux/macOS
- Minimum 8 GB RAM
- Intel Core i5 or higher
- Microphone for recording audio
- Speaker/Headphones for playback

Required Python Packages

Install the required packages using the following command:

pip install -r requirements.txt

The project's requirements.txt contains:

streamlit==1.38.0

openai-whisper==20231117

sentence-transformers==3.0.1

torch==2.3.1

librosa==0.10.2

matplotlib==3.9.0

numpy==1.26.4

soundfile==0.12.1

nltk==3.8.1

fpdf2==2.7.9

pydub==0.25.1

scipy==1.13.1

pandas==2.2.2

Project Setup

Step 1 – Clone the Repository

git clone https://github.com/poshithayagna2007-max/VBCUA.git

Step 2 – Navigate to the Project Folder

cd VBCUA

Step 3 – Create a Virtual Environment

python -m venv venv

Step 4 – Activate the Virtual Environment

Windows

venv\Scripts\activate

Step 5 – Install Dependencies

pip install -r requirements.txt

Step 6 – Run the Application

streamlit run app/app.py

Project Workflow

The development of the **Voice-Based Concept Understanding Analyser (VBCUA)** was divided into six major milestones. Each milestone focuses on implementing a specific functionality required to build the complete AI-powered concept evaluation system.

Milestone 1: Environment Setup and Project Architecture

Activity 1.1: Install Required Software

Before starting development, install the required software and development tools.

- Install **Python 3.10 or later**.
- Install **Visual Studio Code** as the development environment.
- Install **Git** for version control.
- Verify Python installation using:

python --version

Activity 1.2: Create the Project Structure

Create the project folder with separate modules for better maintainability.

```
VBCUA/
│
├── app/
│   └── app.py
│
├── modules/
│   ├── transcription.py
│   ├── semantic_analysis.py
│   ├── audio_features.py
│   ├── scoring.py
│   └── pdf_report.py
│
├── reference_data/
│   └── concepts.json
│
├── assets/
│
├── requirements.txt
│
```

└── README.md

The modular structure improves readability, scalability, and simplifies maintenance.

Activity 1.3: Create Virtual Environment

Create an isolated virtual environment.

python -m venv venv

Activate it.

Windows

venv\Scripts\activate

Install all dependencies.

pip install -r requirements.txt

Activity 1.4: Design the System Architecture

Before coding, the architecture of the application was designed.

The application consists of six major modules:

- User Interface (Streamlit)
- Speech Recognition Module
- Semantic Analysis Module
- Audio Feature Extraction Module
- Scoring Engine
- PDF Report Generator

Each module communicates independently, making the system highly modular and reusable.

Milestone 2: Speech Recognition Module

The objective of this milestone is to convert spoken explanations into text using Artificial Intelligence.

Activity 2.1: Audio Collection

The application allows users to provide explanations using two methods.

- Record directly through the microphone.
- Upload an existing audio file.

Supported formats include:

- WAV

- MP3
- M4A
- OGG

The audio is temporarily stored before processing.

Activity 2.2: Speech-to-Text using OpenAI Whisper

After receiving the audio input, the application uses **OpenAI Whisper** to convert speech into text.

The transcription module performs the following tasks:

- Loads the selected Whisper model.
- Processes the audio file.
- Removes unnecessary silence.
- Generates highly accurate transcription.

Different Whisper models are supported.

- Tiny
- Base
- Small

Users can select the preferred model according to speed and accuracy requirements.

Activity 2.3: Display Transcription

The generated transcription is displayed inside the dashboard.

This enables users to verify whether their speech has been recognized correctly before evaluation begins.

Milestone 3: Semantic Understanding Module

After transcription, the application evaluates conceptual understanding.

Activity 3.1: Load Reference Concepts

Reference explanations and important key points are stored inside

reference_data/concepts.json

Each concept contains:

- Reference Explanation
- Key Points

These are loaded whenever the application starts.

Activity 3.2: Semantic Similarity Analysis

The transcribed explanation is compared with the reference explanation using **Sentence-BERT**.

Sentence-BERT converts both texts into embeddings.

Cosine Similarity is then calculated to determine how closely the student's explanation matches the reference concept.

The similarity score is represented as a percentage.

Higher similarity indicates better conceptual understanding.

Activity 3.3: Key Point Coverage

Apart from semantic similarity, the application also checks whether important concepts have been covered.

Each predefined key point is searched within the student's explanation.

The module identifies:

- Covered Key Points
- Missed Key Points

Coverage Ratio is calculated using

Covered Points

Total Key Points

This metric helps identify missing concepts that the learner failed to explain.

Milestone 4: Audio Feature Analysis

Besides conceptual understanding, effective communication is also evaluated.

Activity 4.1: Audio Processing

The uploaded audio is analyzed using **Librosa**.

The module extracts important speech characteristics.

These include:

- Speaking Rate
- Speech Duration
- Pause Duration
- Pause Ratio

- Silent Segments
- Audio Waveform

Activity 4.2: Filler Word Detection

The application identifies unnecessary filler words such as

- Umm
- Uh
- Like
- Actually
- Basically

Frequent filler words reduce fluency.

The total filler count contributes to the final fluency score.

Activity 4.3: Fluency Score Calculation

Speech quality is evaluated using:

- Speaking Speed
- Pause Ratio
- Filler Words
- Speaking Duration

These parameters are combined to generate a **Fluency Score out of 100**.

Milestone 5: Streamlit Application Development

The fifth milestone focuses on developing a responsive and user-friendly web application using the **Streamlit framework**. The objective is to create an intuitive interface that allows users to interact with the AI-powered evaluation system seamlessly.

Activity 5.1: Designing the User Interface

The application interface was developed using Streamlit's interactive components to provide a clean and organized user experience.

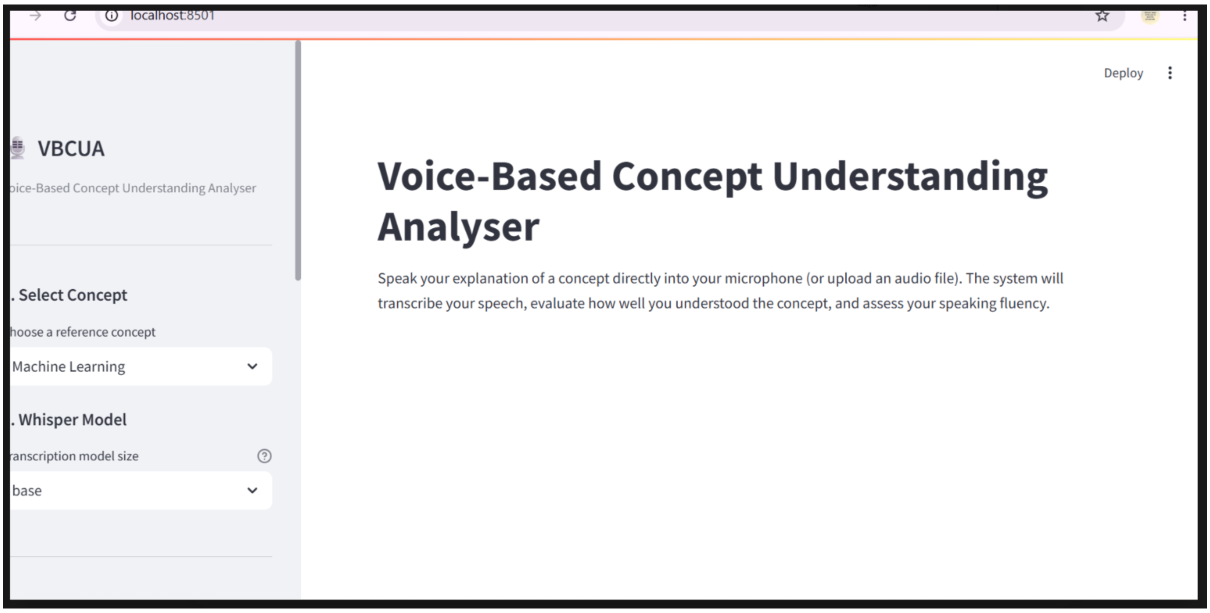
The main interface consists of:

- Project title and description
- Sidebar navigation
- Concept selection
- Whisper model selection

- Audio recording/upload options
- Evaluation dashboard
- Feedback section
- PDF report generation

The layout follows a **wide-screen responsive design**, allowing users to easily access all functionalities from a single page.

Figure 5.1 – Home Page



Activity 5.2: Sidebar Configuration

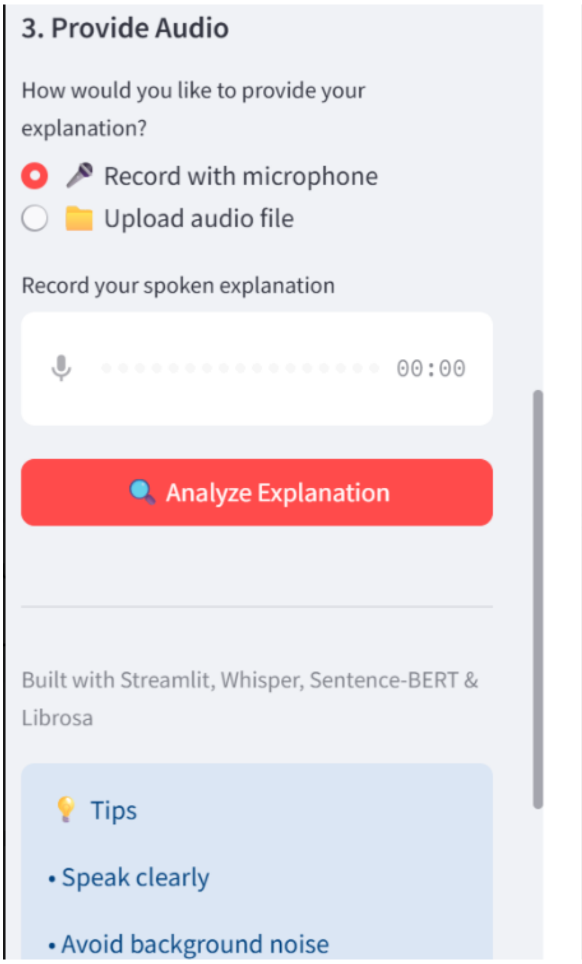
The sidebar acts as the control panel of the application.

Users can:

- Select the reference concept.
- Choose the Whisper transcription model (Tiny, Base, Small).
- Record audio using a microphone.
- Upload audio files.
- Start the evaluation process.

This design minimizes user effort by grouping all input controls into a single location.

Figure 5.2 – Sidebar



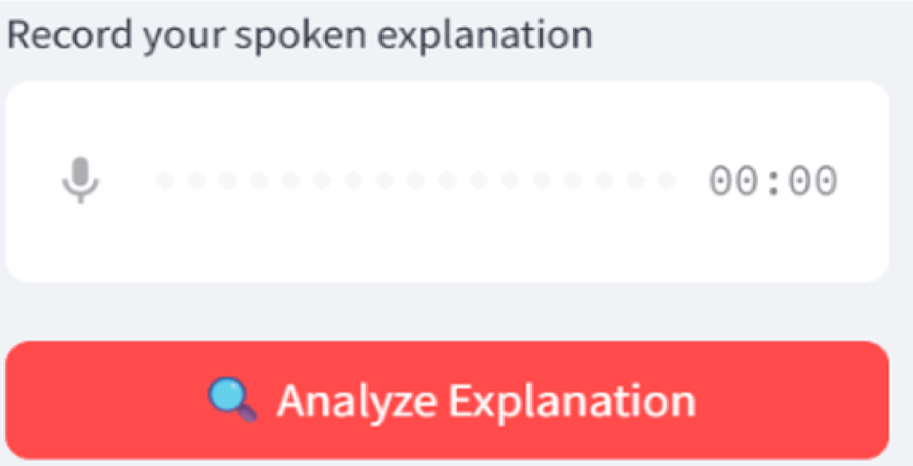
Activity 5.3: Audio Input Processing

The application supports two methods of audio submission:

Microphone Recording

Users can record explanations directly within the application using the built-in Streamlit audio recorder.

Figure 5.3 – Audio Recording Interface



Audio File Upload

Users may also upload previously recorded audio in WAV, MP3, M4A, or OGG format.

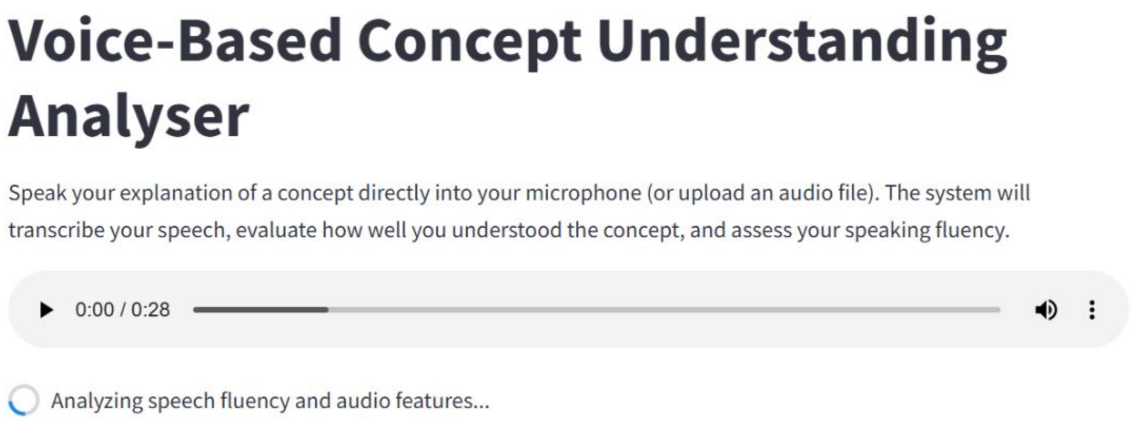
This flexibility enables users to evaluate explanations recorded on different devices.

Activity 5.4: Processing User Input

After the user clicks the **Analyze Explanation** button, the application performs the following sequence:

1. Saves the uploaded audio temporarily.
2. Converts speech into text using Whisper.
3. Computes semantic similarity using Sentence-BERT.
4. Identifies covered and missed concepts.
5. Performs audio feature extraction using Librosa.
6. Calculates fluency score.
7. Computes the overall evaluation score.
8. Generates personalized feedback.
9. Displays the dashboard.

A loading spinner is shown while processing to improve user experience.



Activity 5.5: Evaluation Dashboard

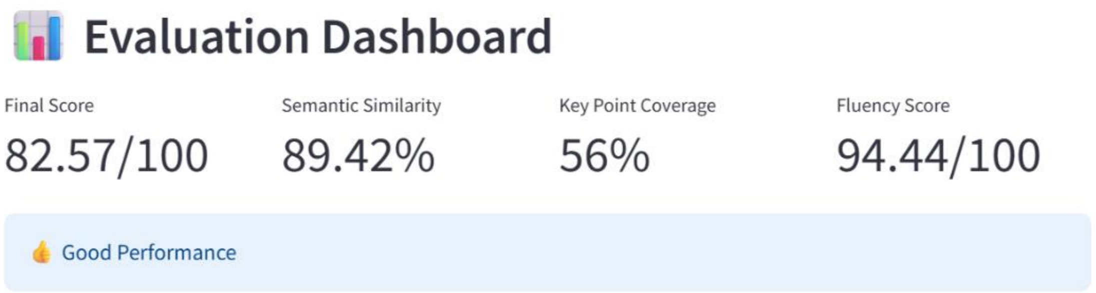
After successful analysis, the application presents the results through an interactive dashboard.

The dashboard displays:

- Final Score
- Semantic Similarity
- Key Point Coverage
- Fluency Score

Users can quickly understand their overall performance from these metrics.

Figure 5.5 – Evaluation Dashboard



Activity 5.6: Performance Visualization

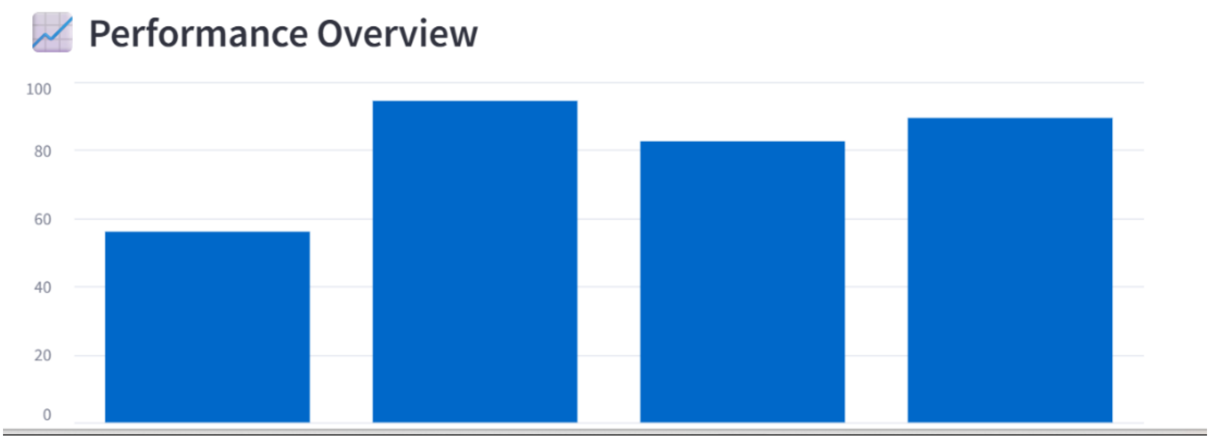
The application provides graphical representations of the evaluation results.

Visual components include:

- Performance bar chart
- Progress indicators
- Individual score bars

These visualizations make interpretation easier for users.

Figure 5.6 – Performance Overview



Activity 5.7: Detailed Result Analysis

The dashboard contains four tabs.

Transcription

Displays the speech converted into text by Whisper.

Semantic Analysis

Displays:

- Covered Key Points
- Missed Key Points
- Coverage Ratio

This helps learners identify concepts that require further study.

Fluency Analysis

Displays:

- Audio waveform
- Speaking rate
- Pause ratio
- Filler words

- Speech duration

These metrics evaluate communication quality.

Feedback

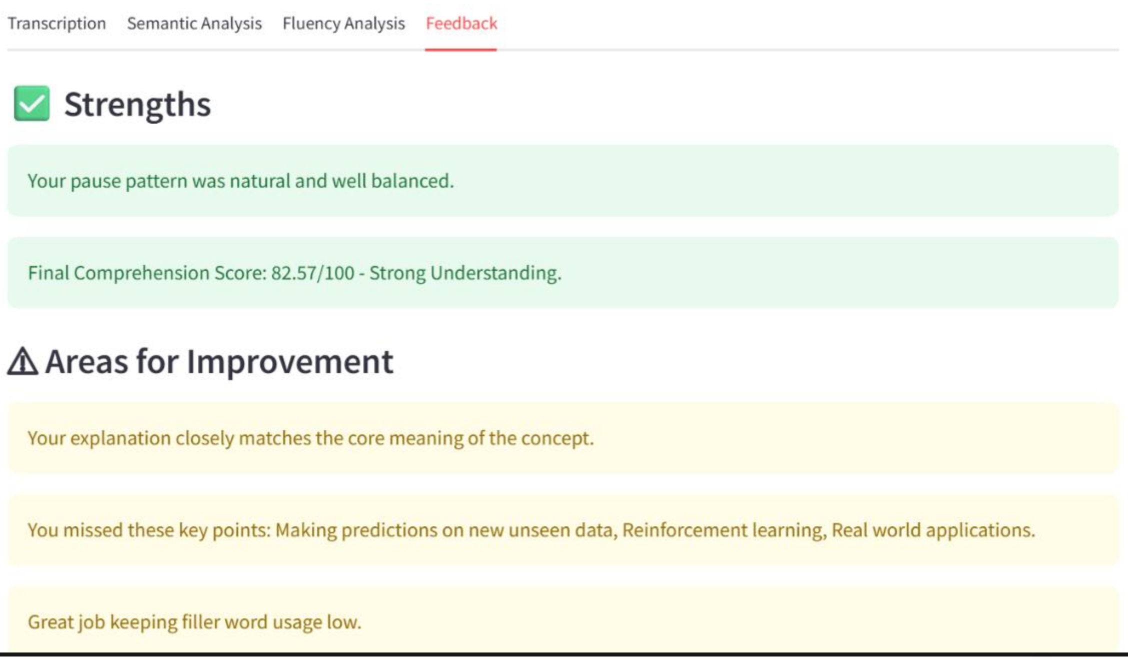
The application generates intelligent feedback based on evaluation results.

Feedback is divided into:

- Strengths
- Areas for Improvement

This enables learners to improve future performance.

Figure 5.10 – Feedback



Activity 5.8: PDF Report Generation

The application automatically generates a detailed PDF report after analysis.

The report contains:

- Concept Name
- Transcription
- Semantic Similarity
- Key Point Coverage
- Fluency Metrics
- Final Score

- Performance Level
- Personalized Feedback

The report can be downloaded using the **Download PDF Report** button.

Milestone 6: Testing and Deployment

The final milestone focuses on validating the application's functionality, verifying the correctness of the AI evaluation process, and deploying the application for practical usage. Comprehensive testing ensures that every module performs accurately and reliably under different user inputs.

Activity 6.1: Functional Testing

The application was tested using multiple spoken explanations across different concepts to verify the correctness of each AI module.

The following functionalities were validated:

- Audio recording through microphone
- Audio file upload
- Speech-to-text conversion
- Semantic similarity calculation
- Key point coverage analysis
- Audio feature extraction
- Fluency score calculation
- Final score generation
- Feedback generation
- PDF report generation

All modules successfully produced the expected outputs.

Activity 6.2: Input Validation

Various input conditions were tested to improve application reliability.

The system correctly handles the following situations:

- No audio uploaded
- Empty microphone recording
- Unsupported audio formats
- Very short recordings

- Long recordings
- Background noise
- Low-quality speech recordings

Appropriate warning and error messages are displayed whenever invalid input is detected.

Activity 6.3: Performance Evaluation

The evaluation system was tested using multiple speech samples with varying levels of conceptual understanding.

The application consistently differentiated between:

- Excellent understanding
- Good understanding
- Average understanding
- Poor understanding

The generated scores closely reflected the quality of the spoken explanations.

Activity 6.4: Deployment

The application can be executed locally using Streamlit.

Execute the following command:

streamlit run app/app.py

The application launches automatically in the default web browser.

Default URL:

http://localhost:8501

The GitHub repository is used for version control and project management.

GitHub Repository

https://github.com/poshithayagna2007-max/VBCUA

Application Screens

This chapter illustrates the major user interface screens of the Voice-Based Concept Understanding Analyser.

Home Page

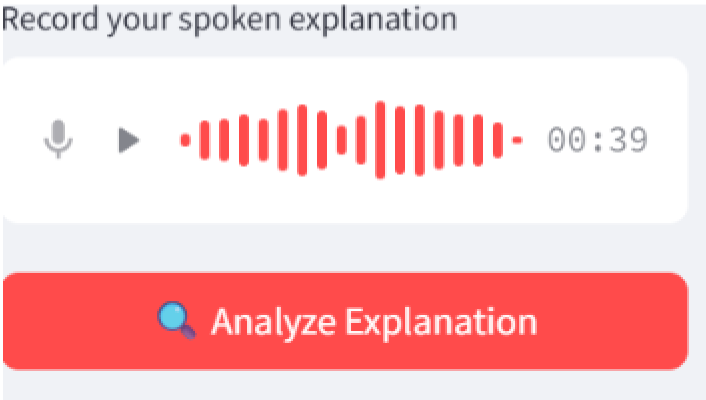
The Home Page provides an overview of the application and introduces users to the purpose of the system. It contains a brief description of the application along with the main evaluation interface.

Sidebar

The sidebar enables users to configure the evaluation process by selecting the reference concept, Whisper model, and preferred audio input method.

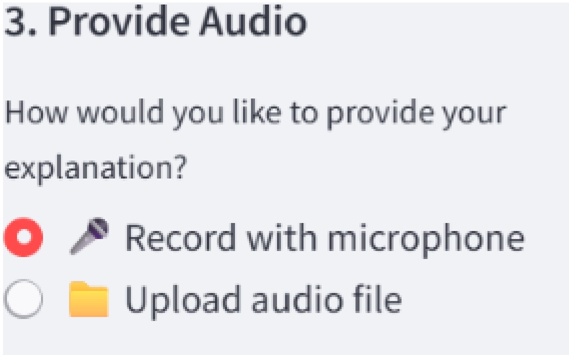
Audio Recording

Users can record spoken explanations directly through their microphone without using external software.



Audio Upload

Users can upload existing recordings in multiple supported audio formats.



Evaluation Dashboard

The dashboard summarizes the evaluation results, displaying semantic similarity, concept coverage, fluency score, and final performance score.

Performance Overview

Performance metrics are presented visually using bar charts and progress indicators, making it easier for users to interpret their results.

Transcription

Displays the speech converted into text by OpenAI Whisper

Semantic Analysis

Displays the covered concepts, missed concepts, and overall concept coverage ratio.

Fluency Analysis

Shows the waveform, speaking rate, pause ratio, speech duration, and filler word statistics extracted using Librosa.

Feedback

Displays personalized strengths and areas for improvement generated from the evaluation process.

PDF Report

Allows users to download a professionally formatted evaluation report containing all assessment details.

Future Enhancements

Although the current version of VBCUA successfully evaluates conceptual understanding through speech analysis, several advanced features can be incorporated in future versions to further improve the system.

Multilingual Support

Support additional languages such as Telugu, Hindi, Tamil, and other regional languages to make the application accessible to a wider audience.

Cloud Deployment

Deploy the application on cloud platforms such as Streamlit Community Cloud, Microsoft Azure, or AWS to enable global accessibility.

User Authentication

Implement secure login functionality allowing students and faculty members to maintain individual profiles and evaluation histories.

Learning Analytics Dashboard

Provide detailed progress tracking, performance trends, and historical reports to monitor learning improvements over time.

Real-Time Evaluation

Enable continuous evaluation during live presentations or classroom activities by processing speech in real time.

Learning Management System (LMS) Integration

Integrate the application with Moodle, Google Classroom, and other LMS platforms for seamless academic assessment.

AI-Based Personalized Recommendations

Recommend study materials and practice questions based on the learner's weak concepts and performance.

Conclusion

Voice-Based Concept Understanding Analyser (VBCUA) is an Artificial Intelligence-powered educational assessment system developed to evaluate conceptual understanding through spoken explanations. By integrating speech recognition, semantic similarity analysis, natural language processing, and audio signal processing, the application provides an objective, efficient, and comprehensive evaluation of a learner's knowledge and communication skills.

The application successfully combines **OpenAI Whisper** for speech transcription, **Sentence-BERT** for semantic similarity analysis, and **Librosa** for fluency assessment to generate an overall performance score along with detailed feedback. The interactive Streamlit interface and automated PDF report generation enhance usability and make the system suitable for educational institutions, online learning platforms, and self-learning environments.

The modular architecture adopted in this project improves maintainability, scalability, and future extensibility. Features such as concept coverage analysis, speech fluency evaluation, graphical performance visualization, and personalized feedback distinguish the system from traditional assessment methods.

Future enhancements including multilingual support, cloud deployment, user authentication, learning analytics, and LMS integration can further transform VBCUA into a comprehensive intelligent learning assessment platform.
Overall, the project demonstrates the practical application of Artificial Intelligence and Machine Learning technologies in modern education by providing a reliable, automated, and data-driven solution for evaluating conceptual understanding through speech.

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